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Impact Statement: Domain-Specific Small Language Model for Clinical Support

Project Objective

This project aims to tackle the significant challenge that healthcare providers and patients encounter when trying to obtain immediate, transparent, and evidence-supported medical guidance. In critical clinical scenarios, the process of analysing huge amounts of intricate literature can cause delays in decision-making, while patients often resort to unreliable health information available online. We suggest developing a specialised small language model (SLM) designed to assess clinical symptoms and consolidate peer-reviewed medical research into practical, evidence-based recommendations.

Benefits

For Healthcare Professionals: Reduces the amount of stress and decreases the time spent on information retrieval, acting as an immediate clinical decision-support tool right at the point of care.

For Patients: Provides straightforward, safe, and comprehensible health information that boosts health literacy and encourages collaborative decision-making.

For the Community: Improves care consistency, minimises triaging bottlenecks, and scales reliable medical information to underserved primary healthcare centres via low-resource SLM deployment.

Potential Risks and Harms

Clinical Misunderstanding: Hallucinations or incorrectly applied evidence could lead to misleading clinical advice, endangering patient safety if not addressed.

Automation Dependence: Excessive trust in automated results may lead users to overlook essential clinical reasoning or to overly rely on the system's judgment.

Demographic and Data Bias: Standard literature sources may fail to adequately represent certain local populations, uncommon conditions, or regional health trends, resulting in varied performance of the model.

Privacy Breaches: Failures in data sanitisation protocols could lead to the risk of re-identifying individuals or exposing sensitive patient information.

Core Values Integrated

Safety & Dignity: Protects human life by ensuring the model is used solely as a support tool grounded in evidence, promoting patient dignity through precise and compassionate care assistance.

Privacy & Transparency: Maintains confidentiality by adhering to stringent data de-identification while offering clear references to source materials, allowing users to verify model outputs.

Inclusivity & Autonomy: Maintains low hardware requirements to ensure accessibility for local users, while safeguarding clinician autonomy by keeping final clinical decisions exclusively in the hands of healthcare professionals.

Mitigation Strategies

Retrieval-Augmented Generation (RAG): Limits model outputs to verified, peer-reviewed literature to avoid inaccuracies and ensure clear source tracking.

Human-in-the-Loop Oversight: Establishes that outputs are solely for decision support, necessitating evaluation by qualified clinicians before any actions are taken.

Rigorous Data Scrubbing: Implements automated, multi-step de-identification processes before handling data to guarantee compliance with privacy regulations.

Clinical Red-Teaming: Performs ongoing validation against established medical standards and multiple peer reviews across specialities to assess accuracy and potential bias.

Limitations and Uncertainties

Evaluating the impact of this project presupposes steady access to current, unbiased medical sources. However, clinical guidelines change frequently, and retrieval systems might not keep pace during new public health developments.